

Problem 1:

1.) $f(x) = x^3 - 0.115x^2 + 3.993 \cdot 10^{-4} = 0$

$x_1 = 0.05$, $x_0 = 0.02$

$f(x_1) = 1.118 \cdot 10^{-4}$

$\frac{f(x_1)(x_1 - x_0)}{f(x_1) - f(x_0)} = \frac{1.118 \cdot 10^{-4} (0.03)}{1.118 \cdot 10^{-4} - 3.334 \cdot 10^{-4}} = -2.183 \cdot 10^{-4}$

(2) $x_{i+1} = x_i - \frac{f(x_i)(x_i - x_{i-1})}{f(x_i) - f(x_{i-1})}$

$f(x_0) = 3.334 \cdot 10^{-4}$

$x_2 = x_1 - \frac{f(x_1)(x_1 - x_0)}{f(x_1) - f(x_0)} = 0.05 + 0.064693$

Error = $|0.064693 - 0.05| = 0.014693$

2.)

Iteration	Approximate root	Error
1	0.064693	0.014693
2	0.057143	0.007263
3	0.058979	0.001549
4	0.058864	0.000115